



Phishing Attack Detecting System

Project Thesis

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Declaration

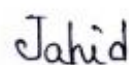
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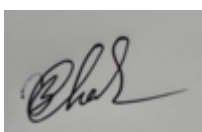
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Approval

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Abstract

The primary purpose of this research is to detect Phishing attacks for secured browsing. Phishing attacks are among the most serious cyber security threats of our time. Phishing is the sending of fraudulent emails, usually intended to deceive recipients into releasing information. There are some significant categories of phishing like Email spoofing, Clone phishing, Spear phishing, Phone phishing, DNS-Based Phishing (Pharming), Man-in-the-middle-attack, and HTTPS phishing, Fast-flux phishing. Phishing attackers can use public information resources, including social networks such as LinkedIn, Facebook, Instagram, and Twitter. Therefore they can gather information about work history, interests, and activities. Our research starts with this concept. First, we have created a filtered website that will detect all fraudulent links and all valid links. We will know about the IP, register date, and name server of all accurate websites with this website.

There is so much research regarding this subject. Most of them have some limitations, and the research is based on some inefficient approaches to overcome this issue. We have tried to solve the problem and practically have a prototype that research is more reliable than others. This is the first try so it will improve with more research. There is a possibility for further development in this project.

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Chapter 1: Introduction

The first phishing attacks began in the mid-1990s. According to the report, phishing attacks were responsible for more than 80% of security incidents in 2020. The 2020 Verizon Data Breach Investigations Report found that 22% of all data breaches involved phishing, and dark web monitoring firm ID Agent estimates that phishing attacks have increased more than 600% since the start of the COVID-19 pandemic [22]. According to the report, the vast majority of these attacks happen via email (96%) [23]. This research also can help to get access to seamless browsing as well as getting access to secured websites. If there are any potential threats are found like unavailability of the webpage domain in all indexes, then the system can identify them. To implement this website-led model, we will build a prototype model that will demonstrate the actuality of the proposal.

1.1 Problem Statement

Our website is able to detect any spam links. It can detect IP address, domain name, register domain id, register IANA id, Domain status, and Name server.

1.1.1 Spam Link Detection

The feature is all about detecting the compromised URL link and sending an alert to the user at a time. If it finds malicious or invalid website threats, the system will detect them. The system evaluates the keywords included in the provided website and thus determines the link's contents.

1.1.2 Details of Valid Website

Link spoofing is an important issue in this digital era. People are constantly being attacked by phishing spoofing. As a result, hackers are easily taking away our personal information. To prevent this attack, we have created a filter website, through which we can understand which website is valid and which website is spamming. Whenever we search for a spoofing link, we can see if this website is under a domain or not registered.

If the link is valid then we may see IP Address, Domain Name, Registry Domain ID, Register URL, Update Date, Creation Date, Registry Expiry Date, Register, Register LANA ID, Register Contact Email, Register Name, Register, Name Server, DNSSEC, and URL of the ICANN Whois Inaccuracy Complaint Form. The domains were sent through the website to obtain the URLs. Approximately half a million unique URLs are crawled first, after which duplicate, and domain-only URLs are removed. After that, the collected URLs were run through the web tool to filter out the safe URLs.

1.1.3 Detecting Created & Expired Websites

We investigate a simple method for detecting and categorizing invalid and genuine websites based on attack type and demonstrate that linguistic analysis is effective and efficient for proactive detection. The biggest advantage is, that when someone sends a spoofing link with the same name then we can also see whether the website register date is valid or not. If it is valid then it will be considered a valid website.

Chapter 2: Theoretical Study

We have taken the help of different languages to create the filtering website. These languages are HTML, BOOTSTRAP, CSS, LARAVEL, PHP, JAVASCRIPT, and JSON.

2.1 HTML (Hypertext Markup Language)

HTML is a markup language used by the browser to manipulate text, images, and other content, in order to display it in the required format. HTML was created by Tim Berners-Lee in 1991. The first-ever version of HTML was HTML 1.0, but the first standard version was HTML 2.0, published in 1999. Hypertext defines the link between the web pages. A markup language is used to define the text document within the tag which defines the structure of web pages. This language is used to annotate text so that a machine can understand it and manipulate text accordingly. Most markup languages are human-readable. The language uses tags to define what manipulation has to be done on the text [32]. It lets the user build and arrange sections, paragraphs, headers, links, and block quotations for websites and apps. HTML is not a programming language, meaning it can't create dynamic functionality. Instead, it makes it possible to organize and format documents, similarly to Microsoft Word, and HTML. Documents are files that end with an HTML extension. We can view them using any web browser (such as Google Chrome, Safari, or Mozilla Firefox). The browser reads the HTML file and renders its content so that internet users can view it [24]. We use HTML for websites Such as homepage, about service, portfolio, contact pages, and resume.

2.1.1 Elements and Tags

HTML uses predefined tags and elements which tell the browser how to properly display the content. Remember to include closing tags. If omitted, the browser applies the effect of the opening tag until the end of the page.

2.1.2 HTML Page Structure

The basic structure of an HTML page is laid out below. It contains the essential building-block elements upon which all web pages are created.

2.2 Bootstrap

Bootstrap is a popular front-end framework for building trendy websites and applications. It's free to use and open-source, however, it comes with many HTML and CSS templates for UI components like buttons and forms. JavaScript extensions are also supported by Bootstrap [25]. We use bootstrap for the front-end development of a web page. It provides a pool of colors, sizes, fonts, and layouts for building our web page. And it enhances the beauty of the web pages.

2.3 CSS (Cascading Style Sheets)

It allows you to create great-looking web pages, but how does it work under the hood? This article explains what CSS is, with a simple syntax example, and also covers some key terms about the language. CSS can be used for even basic document text styling. For example, changing the color and size of headings and links can be used to create a layout. Besides turning a single volume of text into one used for effects such as animation. Have a look at the links in this paragraph for specific examples [26].

CSS, or Cascading Style Sheets, is a primary design language designed to make creating web pages presentable easier. CSS handles the look and feel part of a web page. Using CSS, you can control the color of the text, the style of fonts, the spacing between paragraphs, how columns are sized and laid out, what background images or colors are used, layout designs, and variations in display for different devices, and screen sizes as well as a variety of other effects.

CSS is simple to learn and comprehend, but it gives you a lot of power over how an HTML page looks. CSS is often used in conjunction with the markup languages HTML or XHTML.

2.3.1 CSS Saves Time

You can write CSS once and then reuse the same sheet on multiple HTML pages. You can define a style for each HTML element and apply it to as many Web pages as you want.

2.3.2 Pages Load Faster

If you are using CSS, you do not need to write HTML tag attributes every time. Just write one CSS rule of a tag and apply it to all the occurrences of that tag. So less code means faster download times.

2.3.3 Easy Maintenance

To make a global change, simply change the style, and all elements in all the web pages will be updated automatically.

2.3.4 Superior Styles to HTML

CSS has a much wider array of attributes than HTML, so you can give a far better look to your HTML page in comparison to HTML attributes.

2.3.5 Multiple Device Compatibility

Style sheets allow content to be optimized for more than one type of device. By using the same HTML document, different versions of a website can be presented for handheld devices such as PDAs and cell phones or for printing.

2.3.6 Global Web Standards

Now HTML attributes are being deprecated and it is being recommended to use CSS. So it's a good idea to start using CSS in all the HTML pages to make them compatible with future browsers.

CSS is used to control the style of a web document in a simple and easy way. CSS is the acronym for "Cascading Style Sheet"[33].

2.4 Laravel

Laravel is a web application framework with a syntax that is both expressive and aesthetic. We think development must be a fun and creative experience to be meaningful. Laravel aims to simplify development by simplifying typical chores seen in most online applications, such as authentication, routing, sessions, and caching. Laravel strives to make the development process enjoyable for developers while maintaining application functionality. Happy engineers write the most acceptable code. To that goal, we've endeavored to mix the most significant features of existing web frameworks, such as Ruby on Rails, ASP.NET MVC, Sinatra, and frameworks written

in other languages. Laravel is simple to use while giving strong capabilities for building huge, sophisticated applications. A fantastic container for inversion of control, an expressive migration mechanism, and a tightly integrated testing unit [27].

2.5 PHP

PHP is a widely used server-side programming language that's become increasingly fast and powerful over the years. PHP works well with HTML and databases, making it a great language for anyone interested in building dynamic web applications [28]. PHP began as a tiny open-source project that grew in popularity as more people realized its benefits. In 1994, Rasmus Lerdorf released the initial version of PHP.

- PHP stands for "PHP: Hypertext Preprocessor," a recursive acronym."
- PHP is an HTML-embedded server-side programming language.
- It can handle dynamic content, databases, session monitoring, and even whole e-commerce sites.
- PHP is pleasingly speedy in its execution, mainly built as an Apache module on the Unix side.
- It is integrated with various significant databases, including MySQL, PostgreSQL, Oracle, Sybase, Informix, and Microsoft SQL Server.
- PHP supports a wide variety of significant protocols such as POP3, IMAP, and LDAP. Once launched, the MySQL server performs even the most complex queries with big result sets at record speed. PHP4 included support for Java and distributed object architectures (COM and CORBA), allowing for the first time n-tier programming.

2.5.1 Web Server

PHP is compatible with almost any Web Server software, including Microsoft's Internet Information Server (IIS), however, the Apache Server is the most often used.

2.5.2 Database

PHP is compatible with practically every database software, including Oracle and Sybase, however, the MySQL database is the most often used.

2.5.3 PHP Parser

A parser must be installed to interpret PHP script instructions and create HTML output that can be delivered to a Web browser. This article will show you how to install a PHP parser on your PC.

The main way to store information in the middle of a PHP program is by using a variable.

Here are the most important things to know about variables in PHP.

- All variables in PHP are denoted with a leading dollar sign (\$).
- A variable's value is the result of its most recent assignment.
- Variables are assigned with the = operator, with the variable on the left-hand side and the expression to be evaluated on the right.
- Variables can but do not need, to be declared before assignment.
- Variables in PHP do not have intrinsic types - a variable does not know in advance whether it will be used to store a number or a string of characters.
- Variables used before they are assigned have default values.
- PHP does a good job of automatically converting types from one to another when necessary.
- PHP variables are Perl-like.

PHP has a total of eight data types which we use to construct our variables –

- **Integers** – are whole numbers, without a decimal point, like 4195.
- **Doubles** – are floating-point numbers, like 3.14159 or 49.1.
- **Booleans** – have only two possible values either true or false.
- **NULL** – is a special type that only has one value: NULL.
- **Strings** – are sequences of characters, like 'PHP supports string operations.'
- **Arrays** – are named and indexed collections of other values.
- **Objects** – are instances of programmer-defined classes, which can package up both other kinds of values and functions that are specific to the class.
- **Resources** – are special variables that hold references to resources external to PHP (such as database connections) [34].

2.6 JAVASCRIPT

JavaScript (commonly abbreviated as JS) is a lightweight, interpreted, object-oriented programming language with first-class functions. It's best known as a scripting language for Web pages, but it's also used in various non-browser applications. It's a multi-paradigm, prototype-based scripting language supporting object-oriented, imperative, and functional programming techniques. JavaScript is a client-side scripting language that may be used to create programs that control how web pages react when an event occurs. JavaScript is an easy to learn and also a powerful scripting language, widely used for controlling web page behavior. [29]

Conceived as a server-side language by Brendan Eich (then employed by the Netscape Corporation), JavaScript soon came to Netscape Navigator 2.0 in September 1995. JavaScript enjoyed immediate success and Internet Explorer 3.0 introduced JavaScript support under the name JScript in August 1996. In November 1996, Netscape began working with Ecma International to make JavaScript an industry standard. Since then, the standardized JavaScript is called ECMAScript and specified under ECMA-262, whose latest (twelfth, ES2021) edition is available as of June 2021.

JavaScript is the most popular programming language in the world and that makes it a programmer's great choice. Once you learnt JavaScript, it helps you developing great front-end as well as back-end software's using different JavaScript based frameworks like jQuery, Node.JS etc. JavaScript is everywhere, it comes installed on every modern web browser and so to learn JavaScript you really do not need any special environment setup. For example Chrome, Mozilla Firefox, Safari and every browser you know as of today, supports JavaScript. JavaScript helps you create really beautiful and crazy fast websites. You can develop your website with a console like look and feel and give your users the best Graphical User Experience. JavaScript usage has now extended to mobile app development, desktop app development, and game development. This opens many opportunities for you as JavaScript Programmer. Due to high demand, there is tons of job growth and high pay for those who know JavaScript.

JavaScript, as previously said, is one of the most extensively used programming languages. (Front-end as well as Back-end). It has its presence in almost every area of software development. I'm going to list few of them here:

2.6.1 Client-side validation

This is really important to verify any user input before submitting it to the server and JavaScript plays an important role in validating those inputs at the front-end itself.

2.6.2 Manipulating HTML Pages

JavaScript helps in manipulating HTML page on the fly. This helps in adding and deleting any HTML tag very easily using JavaScript and modify your HTML to change its look and feel based on different devices and requirements.

2.6.3 User Notifications

You can use JavaScript to raise dynamic pop-ups on the webpages to give different types of notifications to your website visitors.

2.6.4 Back-end Data Loading

JavaScript provides Ajax library which helps in loading back-end data while you are doing some other processing. This really gives an amazing experience to your website visitors.

2.6.5 Presentations

JavaScript also provides the facility of creating presentations which gives website look and feel. JavaScript provides RevealJS and BespokeJS libraries to build a web-based slide presentations.

2.6.6 Server Applications

Node JS is based on Chrome's JavaScript engine to create quick and scalable network applications. This is an event-based library that aids in developing advanced server applications, such as Web servers.

JavaScript is a dynamic programming language for computers. It is a lightweight component of web pages whose implementations enable client-side scripts to interact with users and create active sites. It is an object-oriented programming language that is interpreted.

JavaScript was initially known as LiveScript, but due to the buzz around Java, Netscape changed its name to JavaScript. LiveScript was the moniker given to JavaScript when it originally appeared in Netscape 2.0 in 1995. The language's general-purpose core has been integrated into Netscape, Internet Explorer, and other web browsers.

- JavaScript is an interpreted, lightweight programming language.
- Create network-centric apps using this framework.
- Compatible with and complementary to Java.
- Compatible with and complementary to HTML.
- Cross-platform and open
- Client-side, the most extensively used dialect of the language, is JavaScript. The script must be included in or referenced by an HTML page for the code to be interpreted by the browser. It implies that a web page does not have to be static HTML; instead, it might include programs that interact with the user, manage the browser, and dynamically generate HTML content. The JavaScript client-side technique has various advantages compared to conventional CGI server-side scripts. For example, you might use JavaScript to check if a user has entered a valid e-mail address in a form field. When a user submits a form, JavaScript code is executed, and only valid inputs are sent to the Web Server. JavaScript may collect user-initiated events like button clicks, link navigation, and other actions that the user initiates explicitly or indirectly.

2.6.7 Less Server Interaction

Before delivering the page to the server, you may check user input. This reduces server traffic, resulting in lower server strain.

2.6.8 Visitors get immediate feedback

They don't have to wait for the website to refresh to discover whether they've forgotten anything.

2.6.9 Increased Interactivity

You may make interfaces that respond when users move their mouse over them or activate them with the keyboard.

2.6.10 Richer Interfaces

To provide a Rich Interface to your site users, you may utilize JavaScript to incorporate features like drag-and-drop components and sliders. It provides several handy prebuilt JavaScript components, integrates well with databases, and conforms to new standards

such as XHTML and XML. Home Site 5 is a well-liked HTML and JavaScript editor from Macromedia that can be used to manage personal websites effectively [35].

2.7 JSON (JavaScript Object Notation)

The JSON format is syntactically similar to the code for creating JavaScript objects. Because of this, a JavaScript program can easily convert JSON data into JavaScript objects. JSON data may be readily transmitted between computers and utilized by any programming language since it is in text format. [30]

JSON, or JavaScript Object Notation, is a text-based open standard for transferring human-readable data. Douglas Crockford first defined the JSON format detailed in RFC 4627. Application/JSON is the official Internet media type for JSON, and the .json file extension is used for JSON files. This course will help you how to use JSON in various computer languages, including PHP, PERL, Python, Ruby, and Java. JSON, or JavaScript Object Notation, is a text-based open standard for transferring human-readable data. Programmers are familiar with JSON conventions, including C, C++, Java, Python, Perl, and others. [36].

- JSON refers to the JavaScript Object Notation standard.
- Douglas Crockford specified the format.
- It was created to exchange data in a human-readable format.
- The JavaScript programming language has been expanded.
- The .json file extension is used.
- Application/JSON is the JSON Internet Media Type.
- public.json is the Uniform Type Identifier.
- It's used to create JavaScript-based applications, such as browser extensions and webpages.
- JSON is a serialization and transmission standard for structured data across network connections.
- It's generally used to send data from a server to web apps.
- Public data is provided through web services and APIs in JSON format.
- It's compatible with today's programming languages.

Chapter 3: Background Study

Phishing attack is the most common attack in this era. Users have expressed dissatisfaction with the growing technological revolution, which has significantly impacted today's security issues. In recent years, incidents of human information leaking have become more common. [1] In this new era, many security systems are being developed to ensure that security is given top priority and that prevention from being hacked by cyber-criminals is taken into account. Essential prevention is also taken into account to ensure network security is not breached. Employees in the cyber security field are now looking for reliable and consistent detection strategies for phishing websites. [2] Customers confront different security dangers such as cyber offenses due to the widespread use of the internet to execute various activities such as online billing systems, banking transactions, online commerce, password storage, etc. Many cyber-crimes are invisible, such as spam, scam SMS, cyber terrorism, and phishing. Among these cruel assaults, phishing is the most well-known kind of cyber-crime today. [3] According to recent studies, phishing has become one of the most common and most accessible kinds of law-breaking. The frequency of incidents and user vulnerability has increased in recent years, more in tandem with the risk of economic harm. [4] Phishing is an online technique in which dubious methods used by illegitimate authorities collect sensitive data. Email phishing, website phishing, spear phishing, spoofing, Tab off his guard, Evil twin phishing, and other phishing assaults will be used. [6] Phishing is also known as a data breach on a website. [7] Phishing is commonly carried out through email spoofing or texting, and it involves tricking the user into entering personal information on a fake website that looks and feels the same. It aims to address the growing number of phishing attacks by raising client awareness and utilizing various anti-phishing tools. Day by day phishing attacks is increasing. In this paper [8] people are attacked by online phishing links. Users lost their password, bank account, and credit card details information. URL should be less than 56 characters. If the website URL is not HTTP or HTTPS protocol then don't sign-up or log in to your account. Fake websites use @ symbol and special characters. In this paper, they are using Kaggle open-source data set for detecting phishing links. Here SVM, SSL, Feature Vectors algorithm and classification or regression. The phishing problem is extensively present as there are several ways to carry out such an attack,

which implies that one result isn't acceptable to address it. This paper [9] mainly focusing email phishing. They are using Email Authentication Protocols, Email Security Warnings, and Machine Learning Techniques for Detecting Malicious Emails. If someone attack or sends a Phishing link then a machine learning program and email security warning will be shown on the user interface. In this paper [10] they are talking about discussion phishing links. There is no specific solution. In this paper [11] they are using the OTP platform. When phishing link will show the user computer or mobile device. in this paper [12] Phishing is a cybercrime in which emails, telephone, text messages, personally identifiable information, banking details, credit card details, the password is been targeted. Phishing is mainly a form of online identity theft. Social Engineering is being used by the phisher to steal the victim's data and account details. The hacker sends a link via mail to the user of let's say some bank details or any personal information, so now the user goes to that link and fills all the detail in that link, and then the hacker gets all the information of the user. Sometimes they use clone website phishing attacks. So, always check out before visiting any other sites. Look at the domain that should not be modified or changed. In this paper [13] People are facing phishing attacks. This is a common thing. But they are used for the detection of phishing URL Artificial Neural networks and Deep Neural Networks. They are using a dataset that contains 37175 phishing and 36400 legitimate web pages to train the system. In this paper [14] they are talking about the e-banking sector, user financial data is being hacked by hackers. Phishing attacks may include deceptive, malware, and DNS-based attacks. In deceptive attacks, hackers send a deceptive message to bank users. The deceptive message may have some sense of urgency, such as reporting a fake account problem and a fake issue with account activation. The purpose of the deceptive message is to lure the user to interact immediately. In these cases, the phishing attacker is using a sense of urgency to get the attention of the customer by sounding like a report of a serious problem from a trusted source such as PayPal or Citibank. The user then clicks on the link provided to enter the requested financial information. The phisher collects the victim's information because the phisher has full access to the bogus website to which they linked to the user. So do not click any unknown link and go to a browser and type bank website then login account and password then cash out. This method save for the user. Author [15] has made lots of techniques and approaches. The techniques are the Decision tree algorithm, Simhash algorithm, Web logo approach, fuzzy data mining algorithm, machine learning algorithm, TF-IDF information retrieval

algorithm. Which Fuzzy data mining algorithm showed 100% accuracy and the Web logo technique showed 98% accuracy. Some are lacking in this paper which is that no technique can give 100% accuracy. There is also a considerable deficit in their information and evidence. In this paper [16] they talk about more protection against email spoofing in general. We need to keep an eye on the user email password to be encrypted. They also use this authentication protocol to obtain the following security qualities. This paper talk about protection against common blockchain phishing attacks [17]. Blockchain projects are the biggest victims of phishing attacks as a result of increasingly high investment. They have come up with some ways to get protection from phishing attacks. They monitor mail servers, employee, customer, and investor databases, watch the activity on corporate and community sites, and design a DNS alternative in this instance. There are a variety of security measures available to combat Spam issues, but they are not yet fully developed [18]. Many Android apps are also available on the Play Store (Spam Blocker App; Mr. Number - Caller ID & Spam Protection, for example). Using a combination of 30 features, the machine-learning algorithm detected with approximately 99 percent accuracy [19]. And Security Network level protection, authentication, client-side tools, server-side filters and classifiers, and user education are all approaches to phishing attacks. According to the specific analysis, they looked into the use of ANNs for intrusion detection systems and spam filtering for OSNs platforms. [20] Data mining and heuristics, machine learning, and deep learning algorithms are the protection strategies discussed in this paper. Deep learning (DL) for phishing attack detection, Machine learning (ML) for phishing attack detection, Scenario-based phishing attack detection, and Hybrid learning (HL) based phishing attack detection are all discussed in this work.[21] There are lots of available solutions to the users, But we need to introduce more secure ways for the users for detecting valid sites. There was no newer approach to the previous work. We want to propose a more contemporary approach to the phishing detection system to see it by using a filtering website. After reviewing and researching a solution, we think it is the optimum solution for this kind of problem.

3.1 Research Methodology

The processes or strategies used to find, select, process, and analyze information on a subject are referred to as research methodology [31]. We are exposed to cyber-crime on this digital era and phishing is the new tool to breach the sensitive information from the users. By using spam link, hackers can steal the valuable info. This method plays a vital role because our digital footprints are depending on it. Any compromising on digital security can cause various problems. That is why the given web link should be properly checked up every time for ensuring robust security. On the other hand, it is possible to confirm that the system is working properly without any potential error.

Chapter 4: Implementation

Here we use some analytics of a website. In the source of the web checker's (which is being as a web tool) code, you will see the attribute, analytics etc. By using these analytics, It helps to explain the strange behavior of a strange website. Hovering over links without clicking them will reveal their real destination in the status bar of the browser. Web analytics such as bounce rate, page per visit rate, etc. Many parameters are needed when analyzing a phishing attack. We can learn the size of the attack and the target audience in the search results to be made on the mail gateway according to the following parameters.

- Sender's address
- SMTP IP Address(127.0.0.1)
- @letsdefend.io (domain base)
- Let's defend (Besides the Gmail account, an attacker may have sent from the Hotmail account)
- Subject (sender address and SMTP address may be constantly changing)

In the search results, it is important to learn the recipient addresses and time information besides the mail numbers. If harmful e-mails are constantly forwarded to the same users, their e-mail addresses may have leaked in some way and been shared on sites such as Pastebin.

In our model, we use API based approach. It is a fact that emails composed of plain text are boring. For this reason, mail applications provide HTML support, allowing the creation of mails that can attract more attention from users. Of course, this feature has a disadvantage. Attackers can create e-mails with HTML, hiding URL addresses that are harmful behind buttons/texts that seem harmless. If the attacker searches the domain address on the web tool without containing harmful content on it, that address will appear harmless on our web tool, and if it goes unnoticed, you may be mistaken for this address to be harmless. The web tool will give the analysis result.

 .idea	3/23/2022 2:05 AM	File folder
 phishingdetector-app	3/23/2022 12:20 AM	File folder

 app	1/19/2022 7:25 PM	File folder
 bootstrap	1/19/2022 7:25 PM	File folder
 config	1/19/2022 7:25 PM	File folder
 database	1/19/2022 7:25 PM	File folder
 mysql	1/19/2022 7:25 PM	File folder
 nginx	1/19/2022 7:25 PM	File folder
 node_modules	3/23/2022 12:25 AM	File folder
 public	3/23/2022 12:20 AM	File folder
 resources	1/19/2022 7:26 PM	File folder
 routes	1/19/2022 7:26 PM	File folder
 storage	1/19/2022 7:26 PM	File folder
 tests	1/19/2022 7:26 PM	File folder
 vendor	3/23/2022 12:45 AM	File folder

 .editorconfig	1/19/2022 7:25 PM	Editor Config Sour...	1 KB
 .env	1/20/2022 7:48 PM	ENV File	1 KB
 .styleci	1/19/2022 7:25 PM	Yaml Source File	1 KB
 artisan	1/19/2022 7:25 PM	File	2 KB
 composer	3/23/2022 12:45 AM	JSON Source File	2 KB
 composer.lock	3/23/2022 12:45 AM	LOCK File	289 KB
 docker-compose	1/19/2022 7:25 PM	Yaml Source File	2 KB
 Dockerfile	1/19/2022 7:25 PM	File	2 KB
 package	3/23/2022 12:25 AM	JSON Source File	1 KB
 package-lock	3/23/2022 12:25 AM	JSON Source File	709 KB
 phpunit	1/19/2022 7:25 PM	XML Document	2 KB
 README	1/19/2022 7:25 PM	Markdown Source...	5 KB
 server	1/19/2022 7:26 PM	PHP Source File	1 KB
 webpack.mix	1/19/2022 7:26 PM	JavaScript Source ...	1 KB

 Console	1/19/2022 7:25 PM	File folder
 Exceptions	1/19/2022 7:25 PM	File folder
 Http	1/19/2022 7:25 PM	File folder
 Models	1/19/2022 7:25 PM	File folder
 Providers	1/19/2022 7:25 PM	File folder

	app	3/22/2022 7:55 PM	PHP Source File	10 KB
	auth	1/19/2022 7:25 PM	PHP Source File	4 KB
	broadcasting	1/19/2022 7:25 PM	PHP Source File	2 KB
	cache	1/19/2022 7:25 PM	PHP Source File	4 KB
	cors	1/19/2022 7:25 PM	PHP Source File	1 KB
	database	1/19/2022 7:25 PM	PHP Source File	6 KB
	filesystems	1/19/2022 7:25 PM	PHP Source File	3 KB
	hashing	1/19/2022 7:25 PM	PHP Source File	2 KB
	logging	1/19/2022 7:25 PM	PHP Source File	4 KB
	mail	1/19/2022 7:25 PM	PHP Source File	4 KB
	queue	1/19/2022 7:25 PM	PHP Source File	3 KB
	sanctum	1/19/2022 7:25 PM	PHP Source File	3 KB
	services	1/19/2022 7:25 PM	PHP Source File	1 KB
	session	2/20/2022 11:33 PM	PHP Source File	7 KB
	view	1/19/2022 7:25 PM	PHP Source File	2 KB

	factories	1/19/2022 7:25 PM	File folder	
	migrations	1/20/2022 7:02 PM	File folder	
	seeders	1/19/2022 7:25 PM	File folder	

	assets	1/19/2022 7:25 PM	File folder	
	css	3/23/2022 12:20 AM	File folder	
	js	3/23/2022 1:42 AM	File folder	
	.htaccess	1/19/2022 7:25 PM	HTACCESS File	1 KB
	favicon	1/19/2022 7:26 PM	ICO File	0 KB
	index	1/19/2022 7:26 PM	PHP Source File	2 KB
	mix-manifest	3/23/2022 1:01 AM	JSON Source File	1 KB
	robots	1/19/2022 7:26 PM	Text Document	1 KB
	web	1/19/2022 7:26 PM	Configuration Sou...	2 KB

	css	1/19/2022 7:26 PM	File folder	
	js	1/19/2022 7:26 PM	File folder	
	lang	1/19/2022 7:26 PM	File folder	
	views	1/19/2022 7:26 PM	File folder	

 api	1/19/2022 7:26 PM	PHP Source File	1 KB
 channels	1/19/2022 7:26 PM	PHP Source File	1 KB
 console	1/19/2022 7:26 PM	PHP Source File	1 KB
 web	1/19/2022 7:26 PM	PHP Source File	2 KB

Here is the test file which plays vital role. Here we have added array of layer architecture folders like app, bootstarp, config, database, my sql etc.

 Feature	1/19/2022 7:26 PM	File folder	
 Unit	1/19/2022 7:26 PM	File folder	
 CreatesApplication	1/19/2022 7:26 PM	PHP Source File	1 KB
 TestCase	1/19/2022 7:26 PM	PHP Source File	1 KB

Below it is the main homepage of our web tool (screenshot). By searching the valid or invalid address of the mail/link we detected on this web tool, we can see the reputation of the IP address and find out whether it is included in the blacklist as well as much more information from this web checker. The FrontPage is below:



Figure 1.4.1. Index page

In figure 1.1 above we can see the FrontPage of the web tool where we can paste our desired link. On the Domain Information Checker segment, the desired link will be added. Firstly here is the demo of the valid website. For instance facebook.com



Figure 2.4.2. Valid Link Searching

In figure 1.2 there is a screenshot of pasted link. On the Domain Information checker segment, we can see the valid link. When it is detected as a valid website, it will show the domain information.

[←](#)
[→](#)
[↻](#)
127.0.0.1:8000/traced

[Gmail](#)
[YouTube](#)
[Maps](#)

DOMAIN INFORMATION

[Home](#)
[About](#)
[Resume](#)
[Services](#)
[Portfolio](#)
[Contact](#)

[Twitter](#)
[Facebook](#)
[Instagram](#)
[LinkedIn](#)

IP: 157.240.1.35

Domain Name: FACEBOOK.COM

Registry Domain ID: 2320948_DOMAIN_COM-VRSN

Registrar WHOIS Server: whois.registrarsafe.com

Registrar URL: <http://www.registrarsafe.com>

Updated Date: 2022-01-26T16:45:06Z

Creation Date: 1997-03-29T05:00:00Z

Registry Expiry Date: 2031-03-30T04:00:00Z

Registrar: RegistrarSafe LLC

Registrar IANA ID: 3237

Registrar Abuse Contact Email: abusecomplaints@registrarsafe.com

Registrar Abuse Contact Phone: +1-650-308-7004

Domain Status: clientDeleteProhibited <https://icann.org/epp#clientDeleteProhibited>

Domain Status: clientTransferProhibited <https://icann.org/epp#clientTransferProhibited>

Domain Status: clientUpdateProhibited <https://icann.org/epp#clientUpdateProhibited>

Domain Status: serverDeleteProhibited <https://icann.org/epp#serverDeleteProhibited>

Domain Status: serverTransferProhibited <https://icann.org/epp#serverTransferProhibited>

Domain Status: serverUpdateProhibited <https://icann.org/epp#serverUpdateProhibited>

Name Server: A.NS.FACEBOOK.COM
PHISHING LINK DETECTED. IDENTITY THEFT ALEART! .FIREWALL PROTECTED

Name Server: B.NS.FACEBOOK.COM
PHISHING LINK DETECTED. IDENTITY THEFT ALEART! .FIREWALL PROTECTED

Name Server: C.NS.FACEBOOK.COM
PHISHING LINK DETECTED. IDENTITY THEFT ALEART! .FIREWALL PROTECTED

Name Server: D.NS.FACEBOOK.COM
PHISHING LINK DETECTED. IDENTITY THEFT ALEART! .FIREWALL PROTECTED

DNSSEC: unsigned

URL of the ICANN Whois Inaccuracy Complaint Form: <https://www.icann.org/wicf/>

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Designed by SAJJAD

Figure 3.4.3. Valid Link Searching Result

Secondly here is the demo of an invalid website. For instance Facebook.com67f.e33



Figure 4.4.4. Invalid Link Searching

When it is being detected as an invalid website, it will show the domain information. And give a message as an alert like this “IP NOT FOUND OR THEFT ALERT “which mitigates as a successful phishing detection.

It means that the given link breaks the requisites of a valid website. Furthermore, it allows you to set up advanced tracking/generating alerts like tracking, downloads, etc. in just a few clicks.

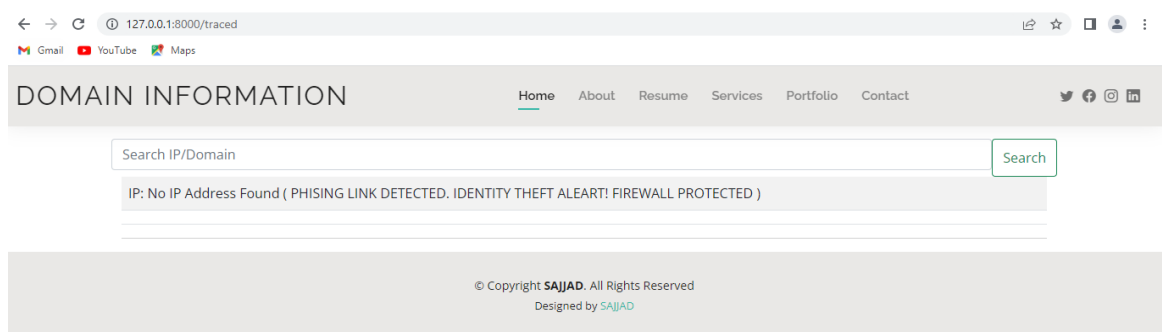


Figure: 4.5. Invalid Link Searching Result

Then we have included some extra features like “CONTACT”. In this way when user faces difficulties, he(s) can communicate through email or phone number as well as using the location. On instant he can get instant reply from developers.

The screenshot shows a web page with a header 'DOMAIN INFORMATION' and a navigation menu with links: Home, About, Resume, Services, Portfolio, and Contact. Social media icons for Twitter, Facebook, Instagram, and LinkedIn are also present. The main section is titled 'CONTACT' and includes a sub-header: 'Communication is a critical skill, as is creating great connections with others in order to get the best outcom'. On the left, there are three contact details: Location (Mirpur,Dhaka,Bangladesh), Email (info@domain.com), and Call (+8809876543212). On the right, there are input fields for 'Your Name', 'Your Email', 'Subject', and a large 'Message' text area. A 'Send Message' button is located at the bottom right of the form.

DOMAIN INFORMATION

Home About Resume Services Portfolio Contact

CONTACT

Communication is a critical skill, as is creating great connections with others in order to get the best outcom

Location: Mirpur,Dhaka,Bangladesh

Email: info@domain.com

Call: +8809876543212

Your Name

Your Email

Subject

Message

Send Message

Figure 5.4.6. Contact Page

Chapter 5: Designs

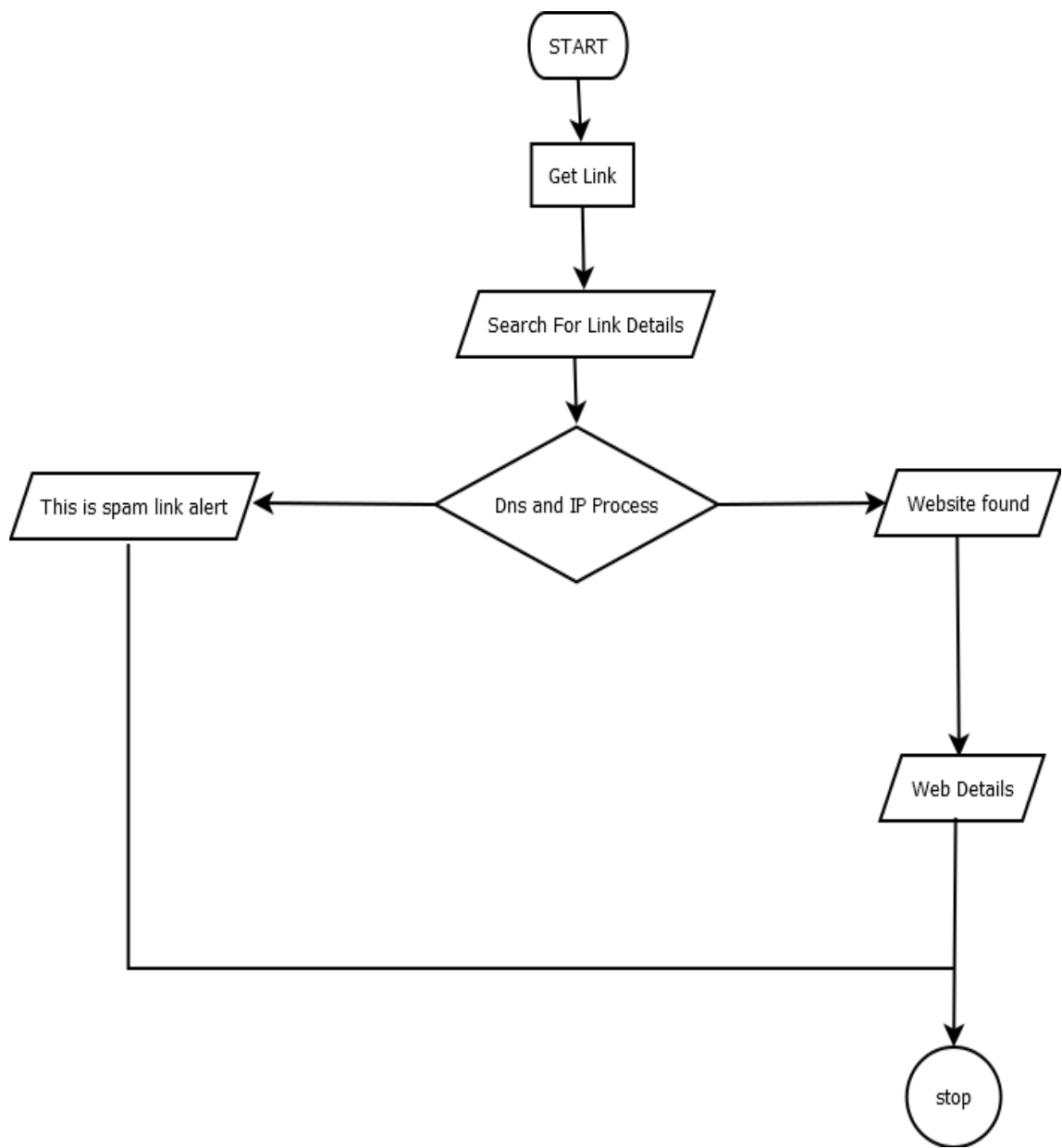


Figure: 5.6. Flow Chart

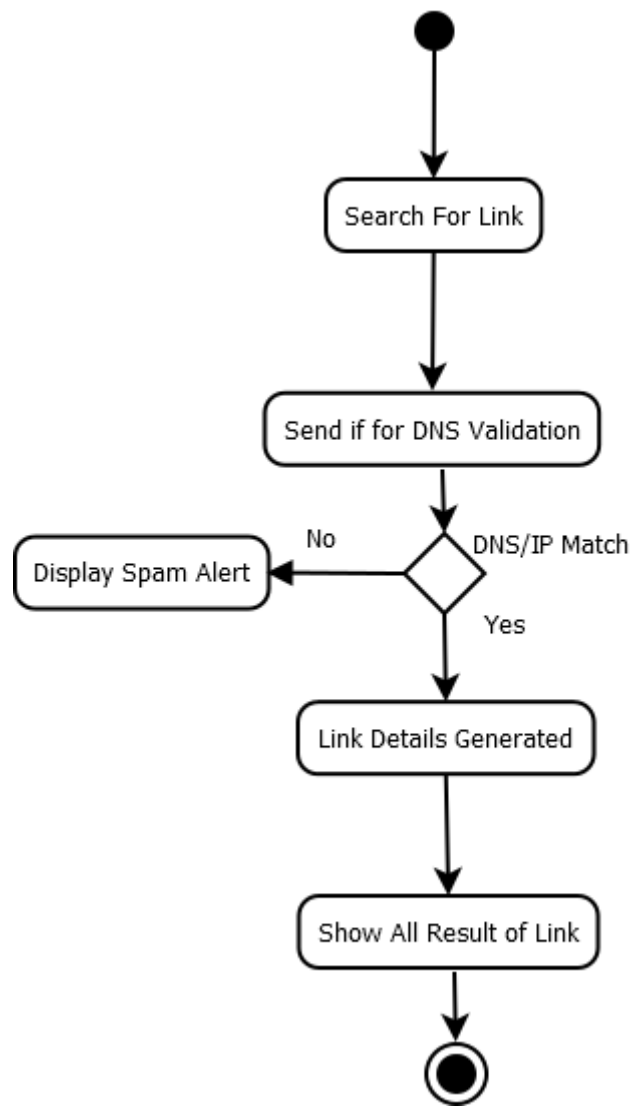


Figure: 5.7. Activity diagram

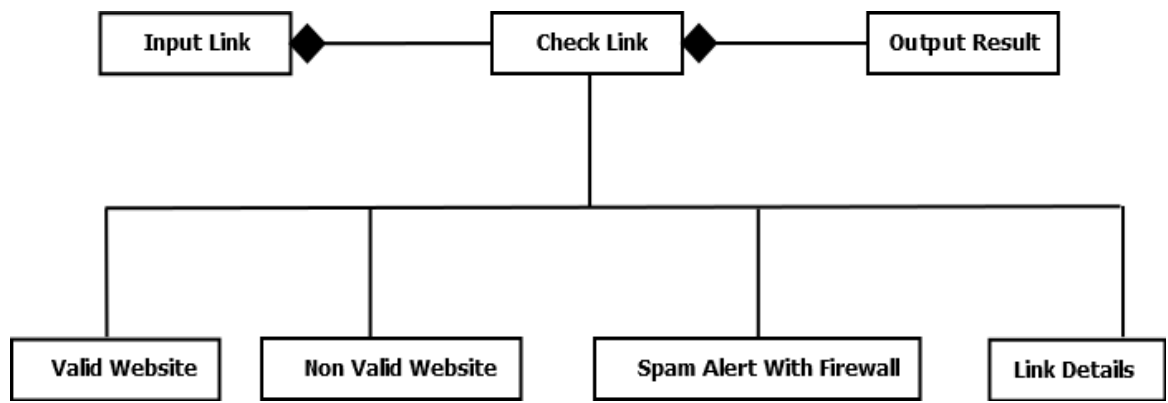


Figure: 5.8. Class diagram

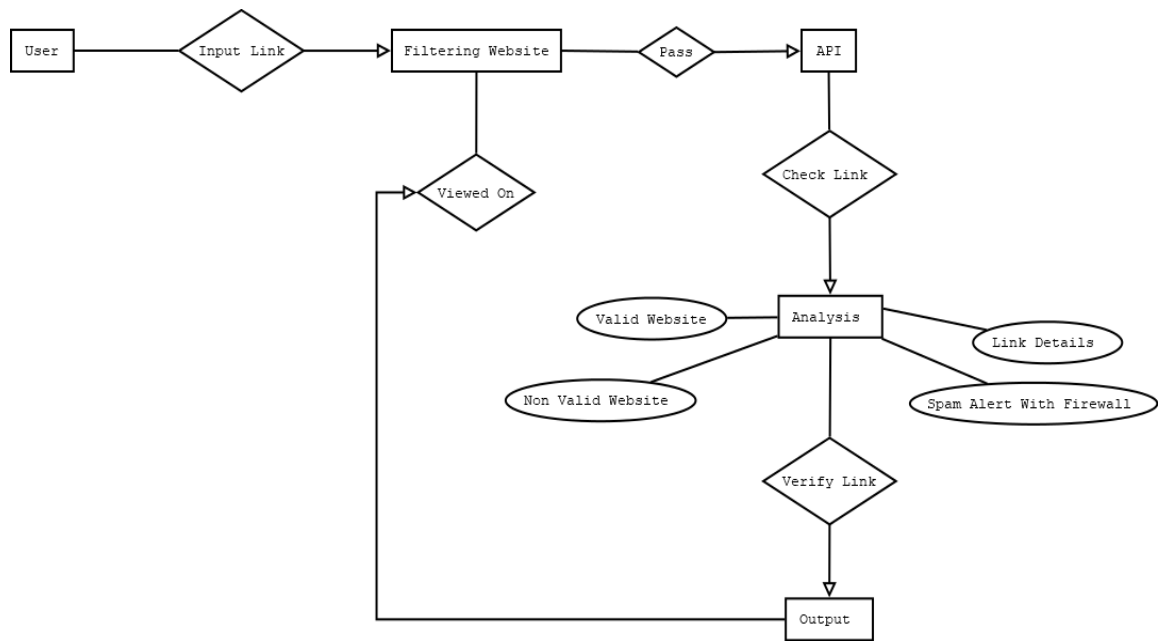


Figure: 5.9. ER diagram

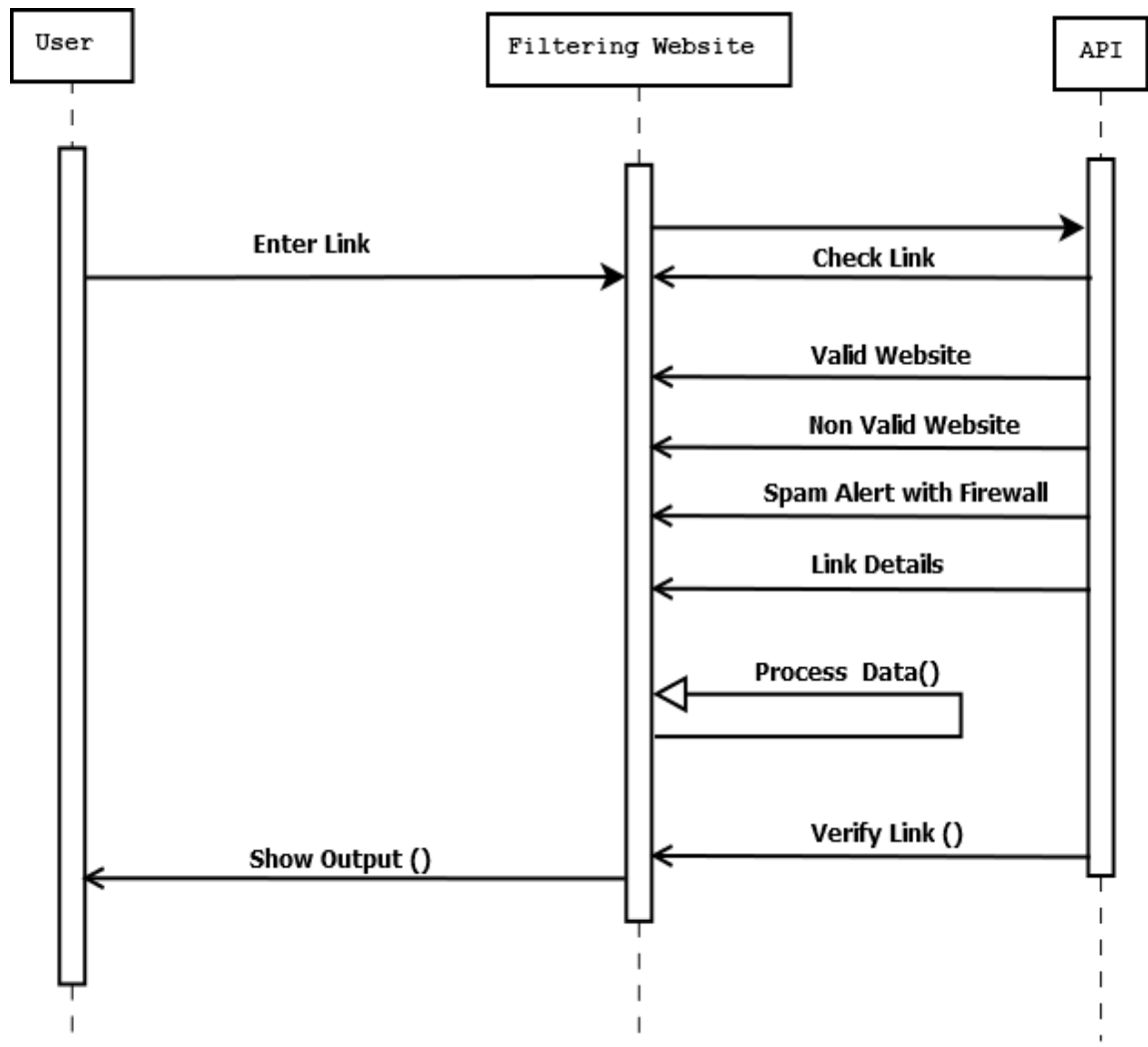


Figure: 10.5. Sequence diagram

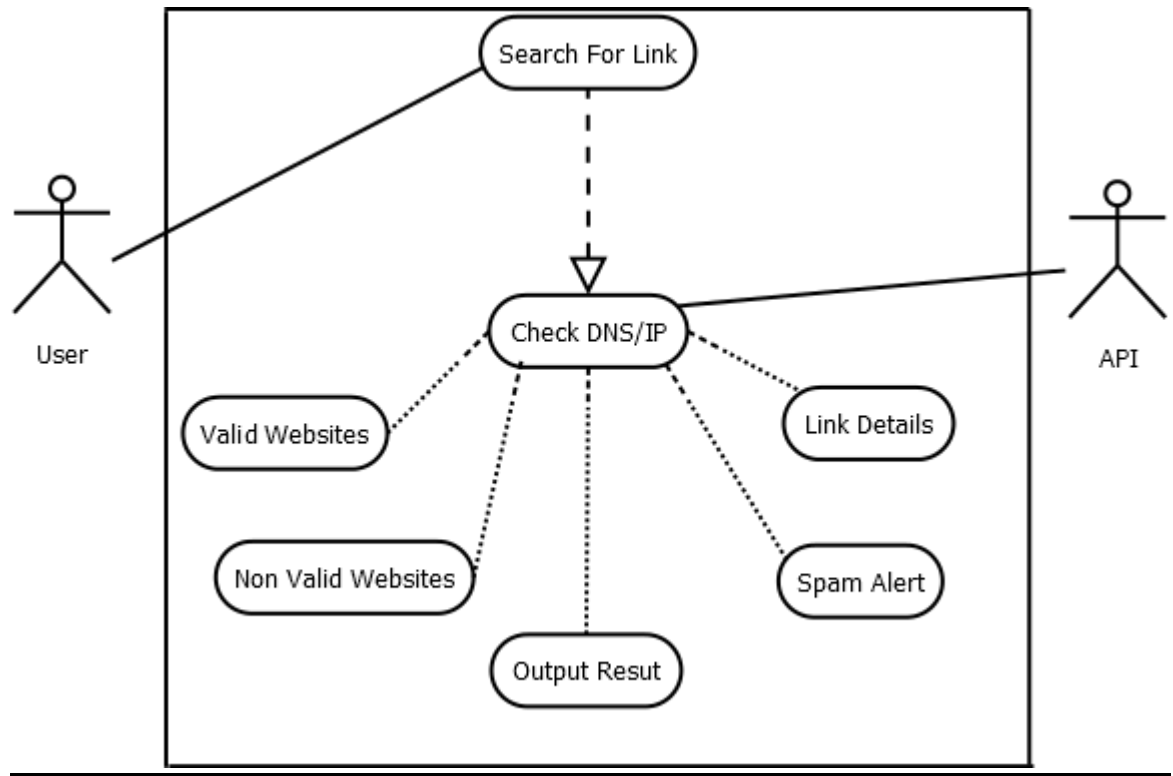


Figure: 5.11. Use Case diagram

Chapter 6: CODES

6.1 Index

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="utf-8">
  <meta content="width=device-width, initial-scale=1.0"
name="viewport">

  <title>Domain Information</title>
  <meta content="" name="description">
  <meta content="" name="keywords">

  <!-- Favicons -->
  <link href="{{asset('assets')}}" /img/favicon.png" rel="icon">
  <link href="{{asset('assets')}}" /img/apple-touch-icon.png" rel="apple-
touch-icon">

  <!-- Google Fonts -->
  <link
href="https://fonts.googleapis.com/css?family=Open+Sans:300,300i,400,40
0i,600,600i,700,700i|Raleway:300,300i,400,400i,500,500i,600,600i,700,70
0i|Poppins:300,300i,400,400i,500,500i,600,600i,700,700i"
rel="stylesheet">

  <!-- Vendor CSS Files -->
  <link href="{{asset('assets')}}" /vendor/aos/aos.css" rel="stylesheet">
  <link
href="{{asset('assets')}}" /vendor/bootstrap/css/bootstrap.min.css"
rel="stylesheet">
  <link href="{{asset('assets')}}" /vendor/bootstrap-icons/bootstrap-
icons.css" rel="stylesheet">
  <link href="{{asset('assets')}}" /boxicons/css/boxicons.min.css"
rel="stylesheet">
  <link
href="{{asset('assets')}}" /vendor/glightbox/css/glightbox.min.css"
rel="stylesheet">
  <link href="{{asset('assets')}}" /vendor/swiper/swiper-bundle.min.css"
rel="stylesheet">
  <!-- Template Main CSS File -->
  <link href="{{asset('assets')}}" /css/style.css" rel="stylesheet">

</head>
```

```

<body>

<x-navbar></x-navbar>

    <!-- ===== Hero Section ===== -->
    <section id="hero" class="d-flex align-items-center">
        <div class="container d-flex flex-column align-items-center" data-
aos="zoom-in" data-aos-delay="100">
            <h1>MD.SAJJAD</h1>
            <h2>Domain Information Checker</h2>
            <form class="form-inline my-2 my-lg-0" method="POST"
action="{{url('/traced')}}">
                @csrf
                <input class="form-control mb-2 mr-sm-2"
type="search" name='search-domain' placeholder="Search
IP/Domain" aria-label="Search">
                <button class="btn btn-outline-success my-2 my-sm-0"
type="submit">Search</button>
            </form>
        </div>
    </section><!-- End Hero -->

    .
    .
    .
    .
    .

    <script
src="{{asset('assets')}}/vendor/glightbox/js/glightbox.min.js"></script>
    >
    <script src="{{asset('assets')}}/vendor/isotope-
layout/isotope.pkgd.min.js"></script>
    <script src="{{asset('assets')}}/vendor/swiper/swiper-
bundle.min.js"></script>
    <script
src="{{asset('assets')}}/vendor/waypoints/noframework.waypoints.js"></s
cript>
    <script src="{{asset('assets')}}/vendor/php-email-
form/validate.js"></script>

    <!-- Template Main JS File -->
    <script src="{{asset('assets')}}/js/main.js"></script>

```

```
</body>

</html>
```

6.2 Welcome

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1,
shrink-to-fit=no">
  <title>Web Tracking</title>
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/css/bootstrap.m
in.css" rel="stylesheet" integrity="sha384-
EVSTQN3/azprG1Anm3QDgpJLIm9Nao0Yz1ztcQTWfSpd3yD65VohhpuuCOmLASjC"
crossorigin="anonymous">
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/js/bootstrap.bun
dle.min.js" integrity="sha384-
MrcW6ZMFYlzcLA8Nl+NtUVF0sA7MsXsP1UyJoMp4YLEuNSfAP+JcXn/tWtIaxVXM"
crossorigin="anonymous"></script>
</head>

<body>
<nav class="navbar navbar-light bg-light">
  <div class="container-fluid">
    <a class="navbar-brand">Web Tracking</a>
    <form class="d-flex" method="POST" action="{{url('/traced')}}">
      @csrf
      <input class="form-control me-2" type="search"
name='search-domain' placeholder="Search IP/Domain" aria-
label="Search">
      <button class="btn btn-outline-success"
type="submit">Search</button>
    </form>
  </div>
</nav>
.
.
.
.
.
.
.
.
```



```

    .
<main role="main">

    </div>

    <hr>

    </div> <!-- /container -->

</main>

<footer class="container">
    <p>&copy; MD Sajjad </p>
</footer>

<!-- Bootstrap core JavaScript
===== -->
<!-- Placed at the end of the document so the pages load faster -->
<script src="https://code.jquery.com/jquery-3.2.1.slim.min.js"
integrity="sha384-
KJ3o2DKtIkVYIK3UENzmM7KCKRr/rE9/Qpg6aAZGJwFDMVNA/GpGFF93hXpG5KkN"
crossorigin="anonymous"></script>
<script>window.jQuery || document.write('<script
src="../../assets/js/vendor/jquery-slim.min.js"></script>')</script>
<script src="../../assets/js/vendor/popper.min.js"></script>
<script src="../../dist/js/bootstrap.min.js"></script>
</body>
</html>

```

6.3 Show

```

<!doctype html>
<html lang="en">
<head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1,
shrink-to-fit=no">
    <title>Web Tracking</title>
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/css/bootstrap.m
in.css" rel="stylesheet" integrity="sha384-
EVSTQN3/azprG1Anm3QDgpJLIm9Nao0Yz1ztcQTWfspd3yD65VohhpuuCOmLASjC"
crossorigin="anonymous">
    <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/js/bootstrap.bun
dle.min.js" integrity="sha384-
MrcW6ZMFYlzcLA8Nl+NtUVF0sA7MsXsP1UyJoMp4YLEuNSfAP+JcXn/tWtIaxVXM"
crossorigin="anonymous"></script>

```

```

</head>

<body>

<nav class="navbar navbar-light bg-light">
  <div class="container-fluid">
    <a class="navbar-brand">Web Tracking</a>
    <form class="d-flex" method="POST" action="{{url('/traced')}}">
      @csrf
      <input class="form-control me-2" type="search"
name='search-domain' placeholder="Search IP/Domain" aria-
label="Search">
      <button class="btn btn-outline-success"
type="submit">Search</button>
    </form>
  </div>
</nav>

<main role="main">

  <!-- Main jumbotron for a primary marketing message or call to
action -->
  <div class="jumbotron">
    <div class="container">
      <h1 class="display-3">Hello, world!</h1>
      <p>This is a template for a simple marketing or
informational website. It includes a large callout called a jumbotron
and three supporting pieces of content. Use it as a starting point to
create something more unique.</p>
      <p><a class="btn btn-primary btn-lg" href="#"
role="button">Learn more &raquo;</a></p>
    </div>
  </div>

  <div class="container">
    <!-- Example row of columns -->

    <hr>

    .
    .
    .

    .
    .
    .
    .
    .
  
```

```

.
.

</div> <!-- /container -->

</main>

<footer class="container">
  <p>&copy; Md Sajjad <p>
</footer>

<!-- Bootstrap core JavaScript
===== -->
<!-- Placed at the end of the document so the pages load faster -->
<script src="https://code.jquery.com/jquery-3.2.1.slim.min.js"
integrity="sha384-
KJ3o2DKtIkVYIK3UENzmM7KCKRr/rE9/Qpg6aAZGJwFDMVNA/GpGFF93hXpG5KkN"
crossorigin="anonymous"></script>
<script>window.jQuery || document.write('<script
src="../../assets/js/vendor/jquery-slim.min.js"></script>')</script>
<script src="../../assets/js/vendor/popper.min.js"></script>
<script src="../../dist/js/bootstrap.min.js"></script>
</body>
</html>

```

6.4 New Show

```

<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="utf-8">
  <meta content="width=device-width, initial-scale=1.0"
name="viewport">

  <title>Domain Information</title>
  <meta content="" name="description">
  <meta content="" name="keywords">

  <!-- Favicons -->
  <link href="{{asset('assets')}}" /img/favicon.png" rel="icon">
  <link href="{{asset('assets')}}" /img/apple-touch-icon.png"
rel="apple-touch-icon">

  <!-- Google Fonts -->
  <link
href="https://fonts.googleapis.com/css?family=Open+Sans:300,300i,400,40

```

```

0i,600,600i,700,700i|Raleway:300,300i,400,400i,500,500i,600,600i,700,700i|Poppins:300,300i,400,400i,500,500i,600,600i,700,700i"
rel="stylesheet">

    <!-- Vendor CSS Files -->
    <link href="{{asset('assets')}}/vendor/aos/aos.css"
rel="stylesheet">
    <link
href="{{asset('assets')}}/vendor/bootstrap/css/bootstrap.min.css"
rel="stylesheet">
    <link href="{{asset('assets')}}/vendor/bootstrap-icons/bootstrap-
icons.css" rel="stylesheet">
    <link href="{{asset('assets')}}/boxicons/css/boxicons.min.css"
rel="stylesheet">
    <link
href="{{asset('assets')}}/vendor/glightbox/css/glightbox.min.css"
rel="stylesheet">
    <link href="{{asset('assets')}}/vendor/swiper/swiper-
bundle.min.css" rel="stylesheet">
    <!-- Template Main CSS File -->
    <link href="{{asset('assets')}}/css/style.css" rel="stylesheet">

</head>

<body>

<x-navbar></x-navbar>

<body>

<div class="jumbotron jumbotron-fluid">
    <p>
        <br>
        <br>
        <br>
    </p>

    <div class="container">
        <form class="d-flex" method="POST" action="{{url('/traced')}}">
            @csrf

            <input class="form-control mb-2 mr-sm-2"
type="search" name='search-domain' placeholder="Search
IP/Domain" aria-label="Search">
            <button class="btn btn-outline-success my-2 my-sm-0"
type="submit">Search</button>

```



```

    </div>
</footer><!-- End Footer -->

<div id="preloader"></div>
<a href="#" class="back-to-top d-flex align-items-center justify-content-center"><i class="bi bi-arrow-up-short"></i></a>

<!-- Vendor JS Files -->
<script
src="{{asset('assets')}}/vendor/purecounter/purecounter.js"></script>
<script src="{{asset('assets')}}/vendor/aos/aos.js"></script>
<script
src="{{asset('assets')}}/vendor/bootstrap/js/bootstrap.bundle.min.js"></script>
<script
src="{{asset('assets')}}/vendor/glightbox/js/glightbox.min.js"></script>
<script src="{{asset('assets')}}/vendor/isotope-layout/isotope.pkgd.min.js"></script>
<script src="{{asset('assets')}}/vendor/swiper/swiper-bundle.min.js"></script>
<script
src="{{asset('assets')}}/vendor/waypoints/noframework.waypoints.js"></script>
<script src="{{asset('assets')}}/vendor/php-email-form/validate.js"></script>

<!-- Template Main JS File -->
<script src="{{asset('assets')}}/js/main.js"></script>

</body>

</html>

```

6.5 Contact

```

<?php
/**
 * Requires the "PHP Email Form" library
 * The "PHP Email Form" library is available only in the pro version
of the template
 * The library should be uploaded to: vendor/php-email-form/php-email-
form.php
 * For more info and help: https://bootstrapmade.com/php-email-form/
 */

// Replace contact@example.com with your real receiving email address

```

```

$receiving_email_address = 'contact@example.com';

if( file_exists($php_email_form = '../assets/vendor/php-email-
form/php-email-form.php' )) {
    include( $php_email_form );
} else {
    die( 'Unable to load the "PHP Email Form" Library!' );
}

$contact = new PHP_Email_Form;
$contact->ajax = true;

$contact->to = $receiving_email_address;
$contact->from_name = $_POST['name'];
$contact->from_email = $_POST['email'];
$contact->subject = $_POST['subject'];

// Uncomment below code if you want to use SMTP to send emails. You
need to enter your correct SMTP credentials
/*
$contact->smtp = array(
    'host' => 'example.com',
    'username' => 'example',
    'password' => 'pass',
    'port' => '587'
);
*/

$contact->add_message( $_POST['name'], 'From' );
$contact->add_message( $_POST['email'], 'Email' );
$contact->add_message( $_POST['message'], 'Message', 10);

echo $contact->send();
?>

```

Chapter 7: Conclusion

This project is based on the daily cyber related malicious attack through phishing link. With the assistance of technology, our online lives are becoming more secure. However, there is no adequate framework to combat this kind of cybercrime. The attacker imitates a genuine person or organization by producing malicious links or other forms of communication that portray them as a legitimate person or business. The primary emphasis of this study is on the topic of technological safety. This may not be the most reliable method, and the offered technique may not be effective against all forms of assaults. However, our study demonstrates that developing more secure systems to avoid all types of assaults is not tricky. On the other side, a future version of this tool may include a pop-up notice option that alerts users to an invalid website. Finally, when a user is subjected to dangerous assaults, this web-based solution will aid them in remaining vigilant.

Chapter 8: Discussion

A phishing Attack detecting System is the solution for preventing misuse of social engineering problems. Every futuristic work can be done with it. There are some laggings on the website. According to the general discussion, the suggested system has been effectively programmed and built, and it has satisfied the expectations set out during the project proposal phase. The system has been successfully created, and the functionalities have also been built based on the test results. Our online lives are not safe, and that is why, to make it safer, we need a passive security system.

Chapter 9: Future Work

9.1 Existing Features

In this project, we use Html, JavaScript, and bootstrap, Laravel, Php, JSON, and CSS languages. Basically, this is a web-based project and we use this website for spam link detection. When we get a link, we have to evaluate whether it is valid or not. If it is invalid, then the web tool will easily detect it. And if it is valid, it will show some extra information as well. Like IP Address, Domain Name, Registry Domain ID, Register URL, Update Date, Creation Date, Registry Expiry Date, Register, Register LANA ID, Register Contact Email, Register Name, Register, Name Server, DNSSEC, and URL of the ICANN Whois Inaccuracy Complaint Form.

9.2 Extra Features

In the future, we want to improve this detection system. Currently, we have to paste the link for detection purposes. In the future, we can build an extension tool by using different programming languages. So that it will easily give us the notification in real-time. And in this way, we will try to build a mobile-based software. So that people can use this software from different platforms like Android, IOS.

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